

Jakub Kisielewski

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EDUCATION

University of Warwick

BSc (Hons) in Computer Science

Coleg Gwent

A-levels in Mathematics, Further Mathematics and Computer Science (AAA)

Coventry, England

September 2024 - June 2027

Crosskeys, Wales

September 2022 - June 2024

TECHNICAL SKILLS

Languages: Java, C, SQL, Python, GDScript

Developer Tools: Git, GitHub, Godot, PostgreSQL, Make

Concepts: Algorithms & Data Structures, Object-Oriented Design, Concurrency, Low-Level Systems

EXPERIENCE

Incoming Software Engineer @ BGC

London, UK | July 2026 – September 2026

Team Member/Cook @ KFC

Risca, Wales | November 2022 – August 2024

- Served orders in high intensity environment, maintaining quality of service by collaborating with team members

PROJECTS

Airport Traffic Model | *Java, JavaFX*

January 2026 – Present

- Collaborating in **team of 6** to build real-time model to analyse aircraft throughput in various airport layouts
- Interpreted client specification to build list of requirements, informing planning and design of system
- Applied **design patterns** such as Factory, MVC and Pipe & Filter to construct an effective system architecture

Crowd Flow | *Python, Streamlit, Numba, NumPy, Librosa*

February 2026 – February 2026

- Implemented predictive framework to help event organisers plan barrier placements to **protect** concert attendees
- Built a custom Verlet Integration physics engine utilising **JIT compilation** to simulate **2000+** agents at 60FPS
- Modeled realistic crowds, using waveform analysis of audio to dynamically modulate agent behaviour in **real-time**
- Reduced crowd pressure by ~60% using a **Monte Carlo Search** algorithm to generate optimal barrier placement

Maze Runner | *GDScript, Godot, Git*

October 2025 – December 2025

- Developed and released a 2D rouge-like game with a **3**-person team, utilising GDDs to align technical milestones
- Defined procedural generation to dynamically generate winding mazes, utilizing randomized pathfinding with **DFS**
- Architected scalable **Entity-Component System** to implement **8** enemy types and various weapon mechanics
- Integrated audio management system, orchestrating over **50+** spatial SFX and music tracks across game states

Intrusion Detection System | *C, POSIX, UNIX APIs*

November 2025 – December 2025

- Created a **multi-threaded** program to detect and prevent multiple network attacks in high-throughput networks
- Established thread-safe system with mutexes/condition variables to prevent race conditions via granular locking
- Utilised **libpcap** to hook into raw network sockets, manually verifying IPv4 packets to identify malicious activity
- Verified **100%** packet capture accuracy during high traffic periods by dynamically reallocating memory

Text Editor | *C, Linux Kernel, GCC*

August 2025 – September 2025

- Constructed a lightweight terminal text editor featuring syntax highlighting, filetype detection, and file persistence
- Engineered low-level, real-time input handling using UNIX system calls (read, write, ioctl, ftruncate, open, close)
- Optimised speed by building efficient data structures and buffered screen rendering leading to **22µs boot time**
- Mastered software engineering principles including modular design, code organisation, and defensive programming

Robot Maze Explorer | *Java*

December 2024

- Produced an optimal pathfinding algorithm for grid-constrained environments by minimising **Manhattan** distance
- Analysed multiple approaches (A*, DFS, BFS, Dijkstra) to optimise time complexity and memory overhead

VOLUNTEERING

University of Warwick Computing Society - Treasurer

March 2025 - Present

- Elected treasurer of the biggest tech society on Warwick campus by election against **4** other candidates
- Accumulated **£8500** in sponsorship money to ensure smooth running of all events for **400+** members
- Coordinating with a team of **30** executive members to calculate budget and organise events in 25/26 academic year

WHACK Hackathon - Co-Lead Organiser

March 2025 - Present

- Raised **£9000** in sponsorship money to successfully run the largest tech event on Warwick campus during 2025
- Acquiring sponsors, planning event logistics, liaising with student union and managing volunteers
- Presently managing and collaborating with organiser team to double event size for 2026